Lab_4_metrics

November 10, 2025

0.1 ** Crash course on complex networks metrics for brain connectivity **

Let's review how to visualize a brain connectome and then do some metrics

The most common tool for connectivity visualization is Brainnetviewer (in Matlab), There are alternative as Nilearn.

For just graphs, NetworkX or others (including Brain Connectivity Toolbox: BCTpy)

For a review of the mathematical background, we address you to Rubinov & Sporns Neuroimage 2010: https://www.sciencedirect.com/science/article/pii/S105381190901074X

Image from Shenoy Handiru et al. HBM 2021:

[1]:
```
# The most
!pip install nilearn


import pandas as pd
from nilearn import plotting

# Brain atlas centroids
coords = pd.read_csv("coords.txt", header=None)
# Adjacency matrix (brain connectivity)
connectome = pd.read_csv("Conn_mat.txt", header=None)
plotting.view_connectome(connectome.values, coords.values, linewidth=5.0,␣
↪node_size=5.0)
```

```
Requirement already satisfied: nilearn in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (0.12.1)
Requirement already satisfied: joblib>=1.2.0 in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from nilearn) (1.4.2)
Requirement already satisfied: lxml in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from nilearn) (5.3.1)
Requirement already satisfied: nibabel>=5.2.0 in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from nilearn) (5.3.2)
Requirement already satisfied: numpy>=1.22.4 in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from nilearn)
(1.26.4)
Requirement already satisfied: packaging in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from nilearn) (23.1)
```

```
Requirement already satisfied: pandas>=2.2.0 in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from nilearn) (2.3.3)
Requirement already satisfied: requests>=2.25.0 in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from nilearn)
(2.31.0)
Requirement already satisfied: scikit-learn>=1.4.0 in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from nilearn) (1.7.2)
Requirement already satisfied: scipy>=1.8.0 in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from nilearn)
(1.15.3)
Requirement already satisfied: typing-extensions>=4.6 in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from
nibabel>=5.2.0->nilearn) (4.13.2)
Requirement already satisfied: python-dateutil>=2.8.2 in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from
pandas>=2.2.0->nilearn) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from
pandas>=2.2.0->nilearn) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from
pandas>=2.2.0->nilearn) (2025.2)
Requirement already satisfied: six>=1.5 in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from python-
dateutil>=2.8.2->pandas>=2.2.0->nilearn) (1.16.0)
Requirement already satisfied: charset-normalizer<4,>=2 in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from
requests>=2.25.0->nilearn) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from
requests>=2.25.0->nilearn) (3.4)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from
requests>=2.25.0->nilearn) (2.1.0)
Requirement already satisfied: certifi>=2017.4.17 in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from
requests>=2.25.0->nilearn) (2025.4.26)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/Users/radek.kawa/miniconda3/lib/python3.12/site-packages (from scikit-
learn>=1.4.0->nilearn) (3.6.0)
```

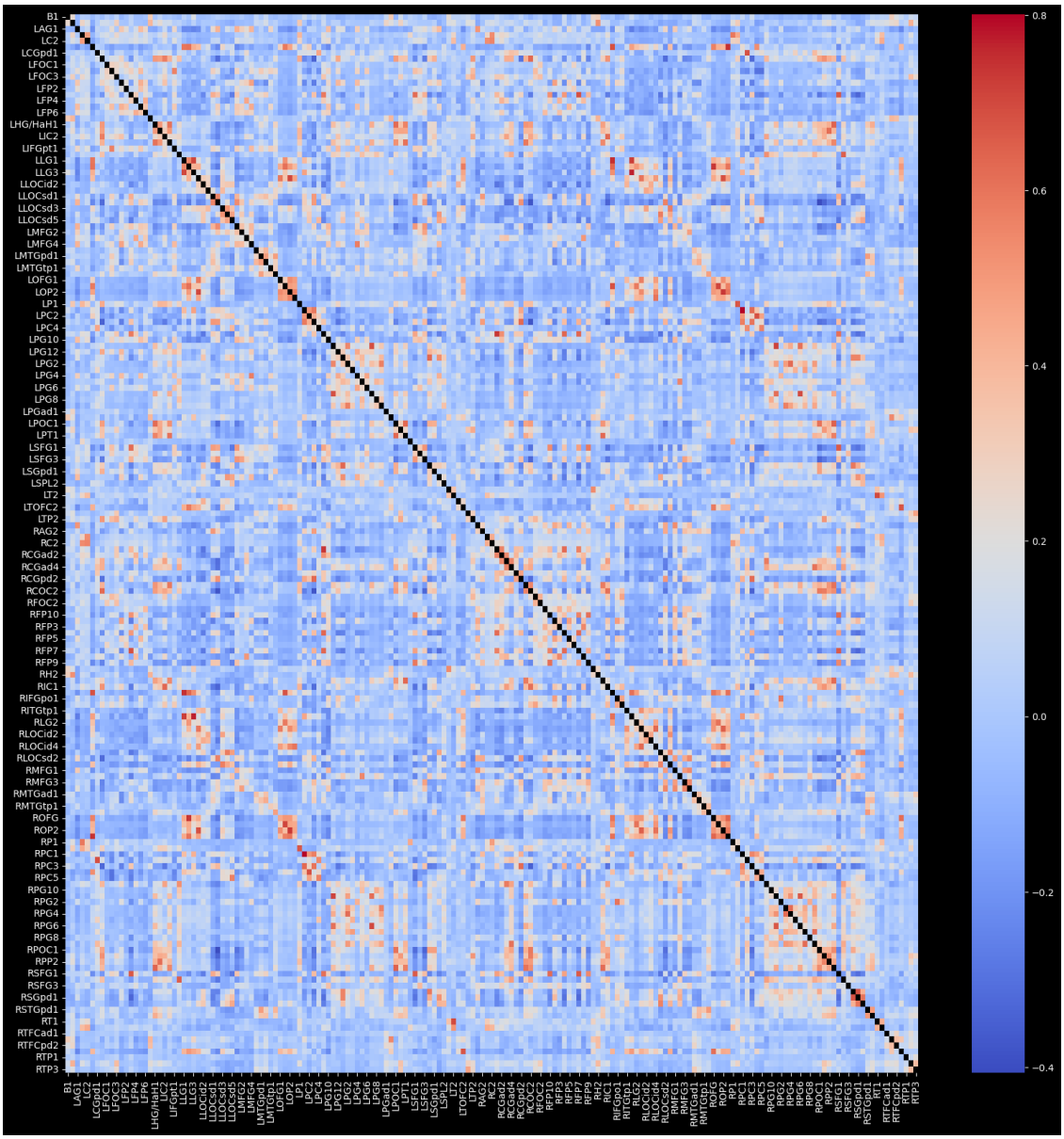
[1]:
```
<nilearn.plotting.html_connectome.ConnectomeView at 0x17439dc10>
```

[2]:
```python
# Import necessary libraries
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
import pandas as pd
import glob

# Network and TDA Libraries
import networkx as nx

#from nxviz import CircosPlot
#import community
```

[3]:
```
#Load complete connectome and names of the ROIs

#load an averaged FUNCTIONAL connectome into matrix
matrix = np.genfromtxt('AveragedMatrix.txt')

#load lables from Glasser atlas
lineList = [line.rstrip('\n') for line in open('ROI_names.txt')]

#Clean the diagonal
matrixdiagNaN = matrix.copy()
np.fill_diagonal(matrixdiagNaN,np.nan)
Pdmatrix = pd.DataFrame(matrixdiagNaN)
Pdmatrix.columns = lineList
Pdmatrix.index = lineList
Pdmatrix = Pdmatrix.sort_index(axis=0).sort_index(axis=1)
```

[4]:
```
#Plot the functional connectome with lables as a heatmap
# This mask variable gives you the possibility to plot only half of the␣
 ↪correlation matrix.
mask = np.zeros_like(Pdmatrix.values, dtype=bool)
mask[np.triu_indices_from(mask)] = True

# This command will show you the functional connectivity matrix with Seaborn
plt.figure(figsize = (20, 20))
_ = sns.heatmap(Pdmatrix, cmap='coolwarm', cbar=True, square=False, mask=None)␣
 ↪# To apply the mask, change to
```



[5]:
```
# Make all values absolute for further user (to simplify our lives in this␣
↪tutorial)
matrix = abs(matrix)
matrixdiagNaN = abs(matrixdiagNaN)


# Most used tools for connectivity metrics are BCTpy and NetworkX

# For NetworkX you need to convert the adjacency matrix in Graph object
# Creating a graph
```

```
G = nx.from_numpy_array(matrix)   #nx.from_numpy_matrix(matrix) for the older␣
 ↪version

# Removing self-loops (is this needed?)
G.remove_edges_from(list(nx.selfloop_edges(G)))
```

[6]:
```
# Review tradional graph metrics

# Computation of nodal degree/strength
#print(nx.degree.__doc__)

strength = G.degree(weight='weight')
strengths = {node: val for (node, val) in strength}
nx.set_node_attributes(G, dict(strength), 'strength') # Add as nodal attribute

# Normalized node strength values 1/N-1
normstrenghts = {node: val * 1/(len(G.nodes)-1) for (node, val) in strength}
nx.set_node_attributes(G, normstrenghts, 'strengthnorm') # Add as nodal␣
 ↪attribute

# Computing the mean degree of the network
normstrengthlist = np.array([val * 1/(len(G.nodes)-1) for (node, val) in␣
 ↪strength])
mean_degree = np.sum(normstrengthlist)/len(G.nodes)
print(mean_degree)
```

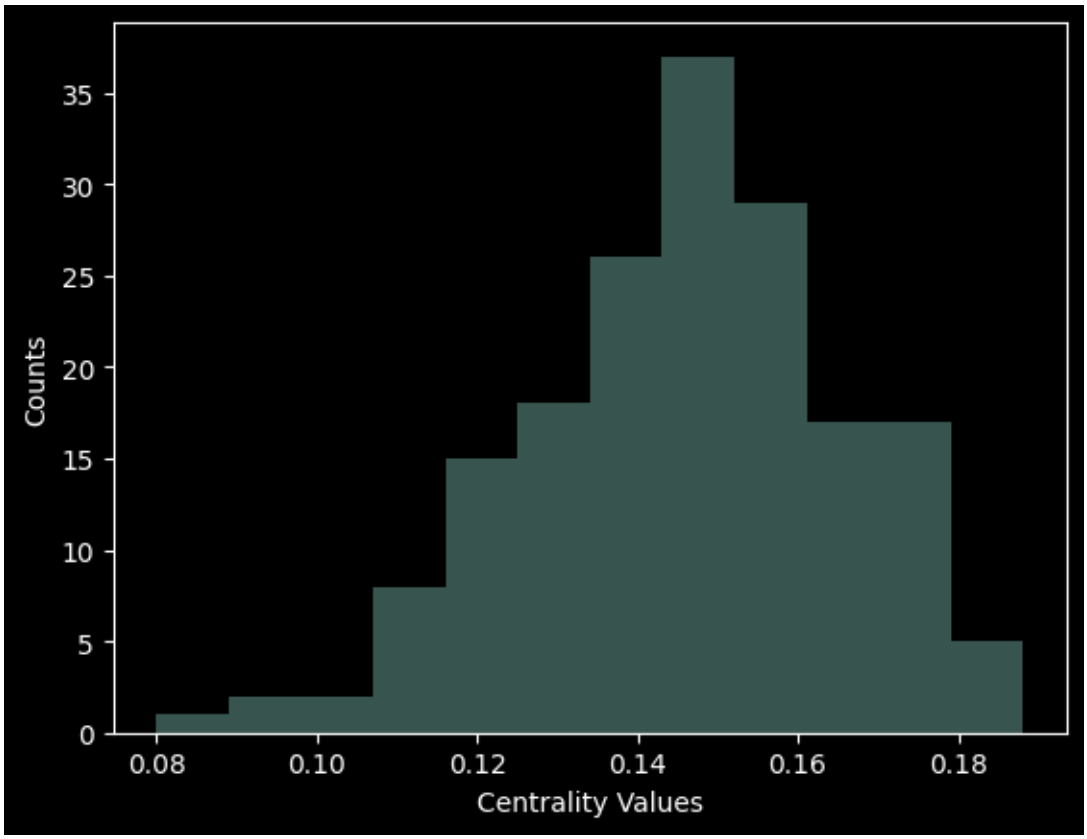
```
0.11425288701460302
```

[7]:
```
# Closeness centrality
#print(nx.closeness_centrality.__doc__)

# The function accepts a argument 'distance' that, in correlation-based␣
 ↪networks, must be seen as the inverse ...
# of the weight value. Thus, a high correlation value (e.g., 0.8) means a␣
 ↪shorter distance (i.e., 0.2).
G_distance_dict = {(e1, e2): 1 / abs(weight) for e1, e2, weight in G.
 ↪edges(data='weight')}

# Then add them as attributes to the graph edges
nx.set_edge_attributes(G, G_distance_dict, 'distance')

# Computation of Closeness Centrality
closeness = nx.closeness_centrality(G, distance='distance')

# Now we add the closeness centrality value as an attribute to the nodes
nx.set_node_attributes(G, closeness, 'closecent')
```

```
# Visualise  values directly
#print(closeness)

# Closeness Centrality Histogram
sns.distplot(list(closeness.values()), kde=False, norm_hist=False)
plt.xlabel('Centrality Values')
plt.ylabel('Counts')
```

```
/var/folders/j8/ycql35hs3415ybxz7vjtcrr40000gn/T/ipykernel_76367/1969302931.py:2
1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(list(closeness.values()), kde=False, norm_hist=False)
```

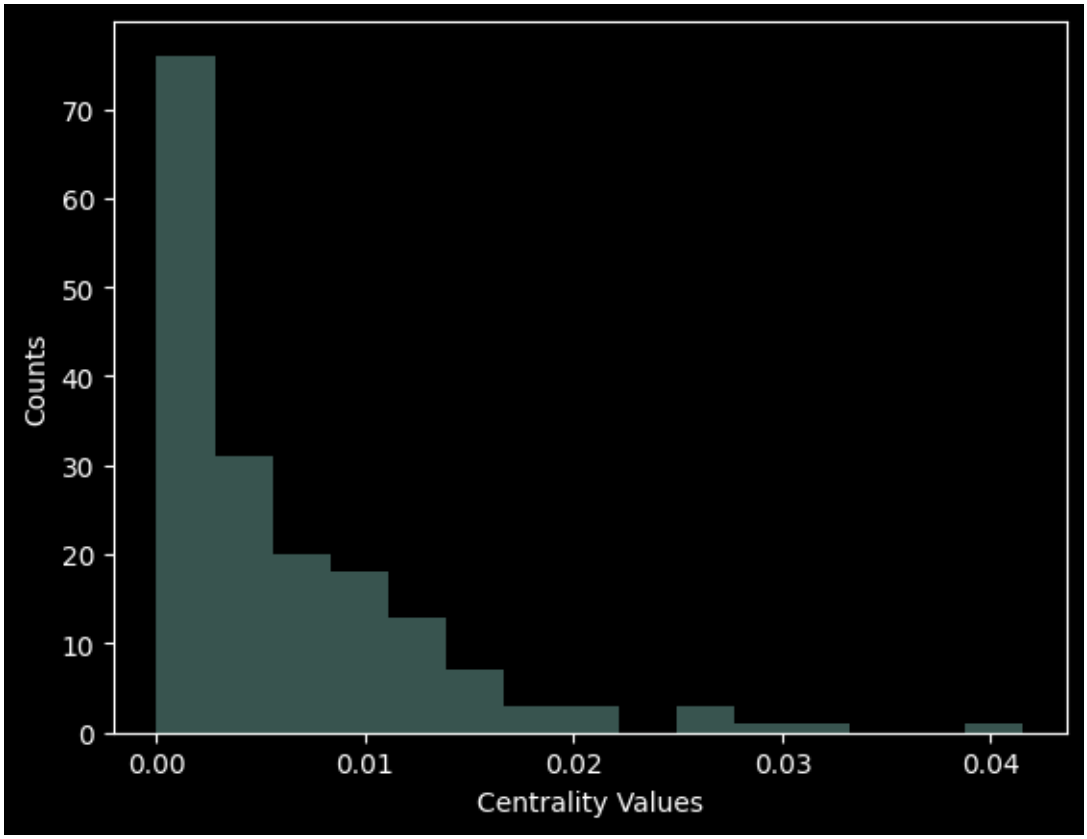
[7]: `Text(0, 0.5, 'Counts')`



[8]:
```
# Betweenness centrality:
#print(nx.betweenness_centrality.__doc__)
betweenness = nx.betweenness_centrality(G, weight='distance', normalized=True)

# Now we add the it as an attribute to the nodes
#nx.set_node_attributes(G, betweenness, 'bc')

# Visualise  values directly
#print(betweenness)

# Betweenness centrality Histogram
sns.distplot(list(betweenness.values()), kde=False, norm_hist=False)
plt.xlabel('Centrality Values')
plt.ylabel('Counts')
```

```
/var/folders/j8/ycql35hs3415ybxz7vjtcrr40000gn/T/ipykernel_76367/3871744565.py:1
2: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(list(betweenness.values()), kde=False, norm_hist=False)
```

[8]:
```
Text(0, 0.5, 'Counts')
```



[9]:
```
# Path Length
#print(nx.shortest_path_length.__doc__)

# This one can also be used if defining source and target:
#print(nx.dijkstra_path_length.__doc__)
nx.dijkstra_path_length(G, source=20, target=25, weight='distance')
```

[9]: 7.81691095077324

[10]:
```
# Clustering Coefficient
#print(nx.clustering.__doc__)
clustering = nx.clustering(G, weight='weight')

# Add as attribute to nodes
nx.set_node_attributes(G, clustering, 'cc')

# Visualise values directly
#print(clustering)

# Clustering Coefficient Histogram
sns.distplot(list(clustering.values()), kde=False, norm_hist=False)
```

```
plt.xlabel('Clustering Coefficient Values')
plt.ylabel('Counts')
```

```
/var/folders/j8/ycql35hs3415ybxz7vjtcrr40000gn/T/ipykernel_76367/2431753096.py:1
2: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(list(clustering.values()), kde=False, norm_hist=False)
```

[10]: `Text(0, 0.5, 'Counts')`